

What Does It Take to Win in the NBA?

A Data-Driven Analysis of Passing and Winning Conditions in Professional Basketball

Emre Ulusoy (s4928539)

Leiden Institute of Advanced Computer Science (LIACS),
Leiden University, The Netherlands
`e.s.ulusoy@umail.leidenuniv.nl`

Abstract. This study looks at a question from Bert Samson, head coach of the basketball club *The Hague Royals*: does more passing in a set offence increase the chance of winning? We use eleven seasons of NBA data (2009–2020), covering 12,777 games and over 2.1 million play-by-play events, and combine hypothesis testing with hypothesis generation to give practical, interpretable answers. We find that the assist differential is the strongest single predictor of winning, that dominating three of four core box-score categories gives an 81% win rate, and that 83% of made three-pointers come directly off an assist. A model using only five features predicts game outcomes with 79% accuracy (AUC 0.87). We also show that although the assist–win link is partly mechanical, a real playmaking effect remains after we control for scoring. The results support the coach’s hypothesis, with some nuance, and are summarised in a one-page coaching dashboard.

Keywords: Sports analytics · Basketball · Subgroup discovery · Predictive modelling · Passing.

1 Introduction

1.1 Client Demand

The client for this project is Bert Samson, head coach of *The Hague Royals*. Coach Samson’s idea is that the more a team passes within a set offence, the more scoring chances it creates, and the more games it wins. His bigger question is simple: “*What do we need to do to win?*” This project turns that practical question into a data analysis and gives back concrete, usable guidance.

1.2 Positioning on the SDS Cube

On the Sports Data Science cube, this project sits clearly: the **sport** is basketball, the **data type** is event and match data, and the **application** is tactics and performance. The work mixes two modes on purpose, since this balance was the main strength noted in our proposal feedback: *hypothesis testing* (checking the coach’s existing ideas) and *hypothesis generation* (finding new winning patterns straight from the data).

1.3 Research Questions

We address three connected questions, derived from the coach’s brief:

- **RQ1 (passing)**: Does more passing, proxied by assists, increase the likelihood of winning?
- **RQ2 (four factors)**: Is it true that dominating four key box-score categories available in our data (assists, rebounds, steals, blocks) effectively guarantees a win?
- **RQ3 (discovery)**: Which specific game conditions are associated with significantly higher or lower win probability?

1.4 Related Work

Analysing basketball with statistics is not new. Oliver’s “Four Factors of Basketball Success” [1] pointed to shooting, turnovers, rebounding and free throws as the main drivers of winning. We borrow the idea of breaking wins down into a few key levers, but it is worth being clear that our four categories (assists, rebounds, steals, blocks) are the ones available in this dataset and are not the same as Oliver’s original four. Interest in ball movement in particular has grown since the three-point revolution of the 2010s. On the method side, subgroup discovery [2] gives us a clean way to do the hypothesis-generation part of this work, since it automatically finds subsets of the data with unusually high or low win rates. What we add is putting these two traditions, the confirmatory and the exploratory, together on one large dataset, in answer to a real coach’s question.

2 Data and Pre-processing

2.1 Dataset

The dataset spans eleven NBA seasons (2009–2020) and consists of three linked tables. The *match* table contains 12,777 games (12,019 regulation, 758 overtime) with per-team statistics. The *player* table provides 351,110 player-game records for 1,468 unique players. The *play-by-play* table records more than 2.1 million timestamped events. Tables are joined via `gameId` and `playerId`.

2.2 Pre-processing

To remove home/away bias and double the effective sample, each game was reshaped into two team-game observations (one per team), yielding 25,554 observations with a balanced 50% baseline win rate. A binary `win` target was derived from the point differential. For each statistical category we computed both absolute values and *differentials* relative to the opponent, the latter proving consistently more informative.

2.3 A Note on Passing Data

One thing shaped the whole analysis: the play-by-play data only records events that end a possession (shots, rebounds, turnovers), not the passes in between. So we cannot count how many passes a team makes in a possession from this data. Because of this, the assist—a pass that leads directly to a made basket—is the best passing proxy we have. We are upfront about this limit rather than claiming the data can show more than it does.

3 Hypothesis Testing

3.1 RQ1: Passing and Winning

Winning teams make on average 24.0 assists per game, against 20.8 for losing teams. The link is highly significant: a point-biserial correlation of $r = 0.31$ ($p < 0.001$) between assists and winning, and a Welch’s t -test confirms the gap (Cohen’s $d = 0.66$, a medium-to-large effect). The *assist differential*, how many more assists a team makes than its opponent, is an even stronger signal ($r = 0.47$). Win rate climbs steadily from 23% for teams with fewer than 15 assists to 81% for teams with 30 or more.

3.2 Possession-Level Evidence

To check the passing effect at the shot level, we parsed the text descriptions of all 374,005 made field goals in the play-by-play data to see which ones were assisted. The result backs up the coach’s idea directly: 83% of made three-pointers came off an assist, against 50% of two-pointers (Figure 1). Across all made shots, 59% were assisted and 41% were not, so unassisted scoring still happens, but it is much rarer on the high-value three-point shots. In other words, high-value outside scoring is almost always the product of ball movement.

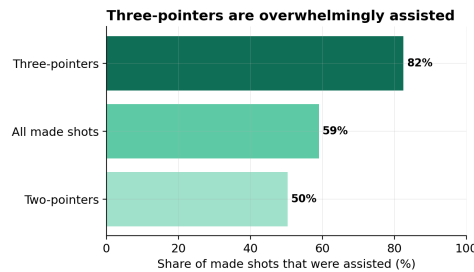


Fig. 1. Share of made shots created by an assist. Three-pointers depend heavily on passing.

3.3 RQ2: The Four-Factor Hypothesis

Next we tested the coach’s idea that dominating the four categories wins games. For each game we counted how many of the four (assists, rebounds, steals, blocks) a team beat its opponent in. A chi-square test showed a strong link (Cramér’s $V = 0.52$, $p < 0.001$). As Figure 2 shows, the pattern is striking: dominating none of the four gives a 9% win rate, three of four gives 81%, and all four gives 95%. Looking at the categories one by one through odds ratios, assist dominance is the strongest single factor (OR = 5.7), ahead of rebounds (3.6), blocks (2.3) and steals (2.0). It is worth flagging that blocks are the weakest of the four. Coach Samson raised exactly this point: with only about 4.9 blocks per game on average (and a difference of two or fewer between the teams in more than half of all games), blocks are too rare to swing a result on their own over 80–90 possessions. The data agrees, and this is reflected later in the low importance blocks carry in the predictive model. We keep blocks in for completeness as one of the available box-score categories, but the real story is assists and rebounds.

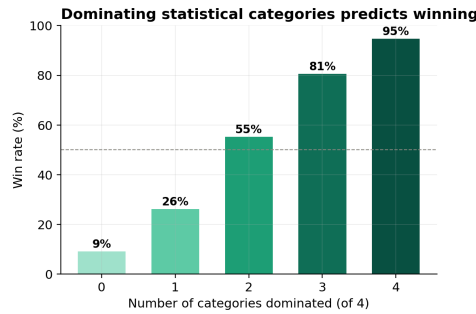


Fig. 2. Win rate by number of statistical categories dominated. Dominating three of four already secures an 81% win rate.

3.4 Controlling for Circularity

There is an obvious catch here. An assist is, by definition, tied to a made basket, a made basket gives points, and points decide wins. So part of the assist–win link could just be “assists mean baskets” in disguise. We tested this head-on. When we control for a team’s own points scored, the raw correlation of $r = 0.31$ drops to a partial correlation of $r = 0.08$ (Figure 4, left), so roughly three-quarters of the raw effect is really just scoring. But the effect does not disappear. In close games (margin ≤ 5 points) the assist differential still separates winners from losers ($r = 0.12$); winning teams turn a higher share of their baskets into assists; and the assist differential stays highly significant when we put it in a model next to the rebound differential, which is not circular (OR = 1.19 per assist).

Our read is that the passing effect is partly mechanical, but a real, independent playmaking effect remains on top of it. Put simply: even when two teams score the same number of points, the team that got there through more passing tends to win a little more often.

4 Hypothesis Generation

4.1 Subgroup Discovery

To find winning conditions we did not specify in advance, we ran subgroup discovery (beam search, WRAcc quality measure) on the binary win target. The strongest pattern it found combines two things: teams with both much higher assists *and* much higher rebounds than their opponent win 94% of games, more than either factor on its own (Figure 3). The flip side is just as clear: teams with both much lower assists and rebounds win only 8%. Notably, the algorithm put assists at the centre of almost every high-win-rate group on its own, which backs up the hypothesis-testing results through a completely separate method.

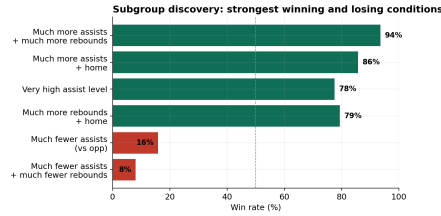


Fig. 3. Conditions discovered by subgroup discovery. Combining high assists with high rebounds yields the strongest winning condition (94%); the inverse yields the weakest (8%).

4.2 Momentum and Timing

Using the quarter-by-quarter scores, we looked at when games are actually decided. The third-quarter margin lines up most strongly with the final result ($r = 0.47$), more than any other quarter (Figure 4, right), which is real evidence for the old basketball saying about the “championship quarter.” A comeback analysis is directly useful for the coach: teams down by up to 5 points at half-time still win 49% of the time, but once they trail by more than 10 that drops to 14%. And in games level after three quarters, the team that wins the fourth wins the game 90% of the time.

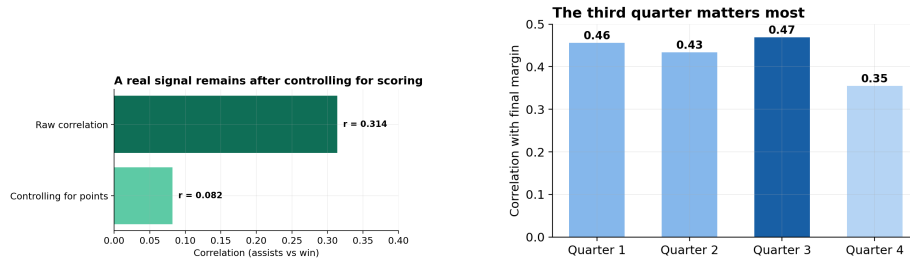


Fig. 4. Left: the assist-win correlation shrinks but persists after controlling for points scored, indicating an independent playmaking signal. Right: correlation between each quarter's margin and the final result; the third quarter is the most decisive period.

4.3 Player Roles and Ball Distribution

Bringing in the player table, we found that guards make 57% of all assists even though they are only about a third of the line-ups, so playmaking is very much a position-specific job. More important for the coach's philosophy is how the ball is shared: teams where nine or more players record an assist win 56% of games, against 41% when only four or fewer do (Figure 5). So it is not just passing more that helps, but spreading the ball around the whole team.

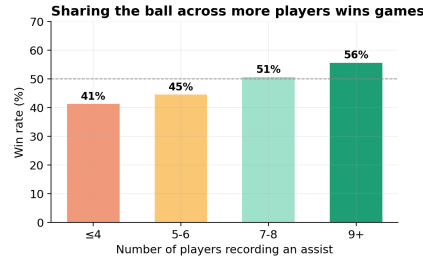


Fig. 5. Win rate rises with the number of players contributing assists, supporting a shared-ball offensive philosophy.

5 Predictive Modelling

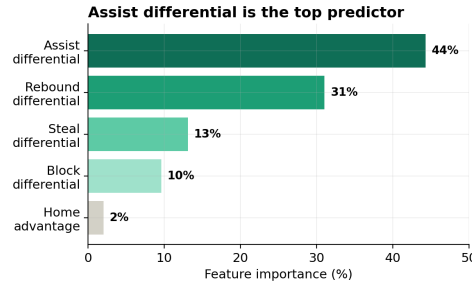
We trained three classifiers to predict the game result from five features: the four statistical differentials plus home advantage. We evaluated them with 5-fold stratified cross-validation and a held-out 25% test set. Table 1 shows the results. Logistic regression did best (79% accuracy, AUC 0.87), matching the much more complex random forest. That the simplest model wins tells us the

Table 1. Model comparison. All three classifiers use the same five features.

Model	CV accuracy	Test accuracy	Test AUC
Logistic regression	0.79	0.79	0.87
Random forest	0.78	0.79	0.87
Decision tree (depth 4)	0.76	0.76	0.81

relationship is largely linear, which also makes it easy to explain to a coach: each extra point of assist differential multiplies the odds of winning by 1.19.

The random forest’s feature importance (Figure 6) sums up the whole project in one chart: the assist differential (44%) is by far the most important predictor, with the rebound differential (31%) next, and steals, blocks and home advantage well behind. Home advantage is almost nothing (2%), which says winning is about *how* a team plays, not where it plays.

**Fig. 6.** Feature importance from the random forest. Assist differential dominates.

6 Robustness

The assist–win link holds up across all eleven seasons (per-season r between 0.38 and 0.51), so it is not a quirk of one particular era. Average assists rose from 21.0 in 2009 to 24.3 in 2020, in line with the league-wide three-point revolution. The link only weakens in overtime games ($r = 0.07$), which makes sense: games close enough to reach overtime are decided by small, fairly random margins rather than by who moved the ball better.

7 Actionable Insights for the Coach

The findings turn into a short, prioritised set of recommendations, gathered into a one-page dashboard for Coach Samson:

1. **Win the assist battle first.** It is the strongest single lever, both in the numbers and on the court.
2. **Pair passing with rebounding.** Together they give a 94% win rate.
3. **Share the ball.** Get nine or more players involved in assists instead of running everything through one or two.
4. **Protect the third quarter.** It is the most decisive period; don't fall more than 10 behind by half-time.

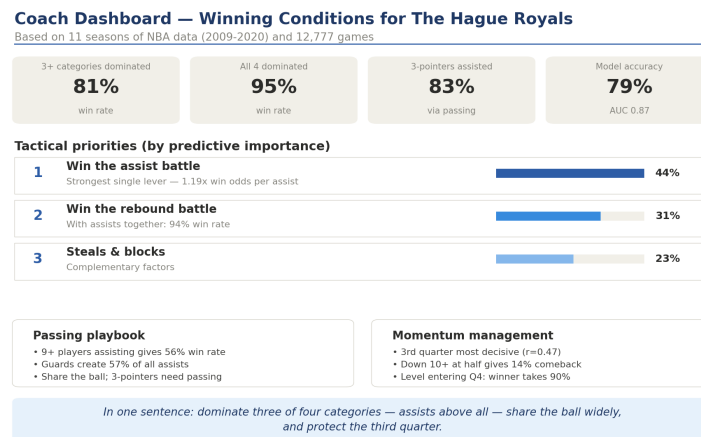


Fig. 7. The one-page coaching dashboard delivered to Coach Samson, summarising the key winning conditions and tactical priorities in an interpretable format.

The dashboard in Figure 7 pulls these findings together into one reference the coach can actually use in training and game prep.

8 Coach Feedback

Because Coach Samson is a real client, we shared the findings with him directly and asked for his reaction. His response added value in several ways. First, he confirmed the headline result from a coaching standpoint, and gave a neat mechanical reason for why passing beats dribbling that the numbers alone do not show: a defender can never keep up with a pass the way they can with a dribble, so ball movement repeatedly creates the open looks our shot-level analysis picked up. Second, he challenged the inclusion of blocks, pointing out that his own teams record only five or six per game, too few to decide a result. Checking this in the NBA data confirmed his intuition (an average of 4.9 blocks per game, with a two-or-fewer difference in over half of games), which is consistent with blocks being the weakest of our four factors.

He also offered two sharper analytical points that are worth recording. He noted that assists and shooting are not independent: shooting well creates more assist chances, so part of what the assist measure captures is simply good shooting. This is a useful framing of the same circularity we test in Section 3.4 from a coach’s perspective. He made a similar point about rebounding, suggesting it is best understood through possessions (shots + turnovers + trips to the free-throw line), since the team with more possessions tends to rebound better. Both remarks point to good directions for future work using possession-level data.

Finally, he asked to use the material with his team and regional union next season, which is the clearest sign the work meets the brief of being practical and usable.

9 Limitations

Two limits are worth stating plainly. First, the data has no real pass sequences, only events that end a possession, so we could never measure passes-per-possession directly and had to rely on assists as a proxy. The swing-pass dynamics the coach originally asked about are simply not in this dataset. Second, part of the assist–win link is mechanical, since assists are tied to made baskets by definition; we addressed this by controlling for scoring, but we cannot remove it entirely. The assist effect also breaks down in overtime games. We report these openly so the findings are read for what they are.

10 Conclusion

Coach Samson’s hypothesis holds up: passing, measured through assists, is strongly linked to winning, and it is the most important of the four box-score factors we looked at. Three separate methods—hypothesis testing, subgroup discovery and predictive modelling—all point the same way, which gives us confidence in the result. Coach Samson’s own feedback adds a fourth, practical layer of validation, including a useful mechanical explanation for why passing works (a defender cannot keep up with a pass as they can with a dribble). The more careful conclusion is that ball movement matters *beyond* just scoring more: teams that produce the same points through more shared, better-quality shots still win a bit more often. With player-tracking data, future work could count passes per possession directly and dig into the swing-pass question the coach raised.

Code and Reproducibility

All analysis code and figures are available at: <https://github.com/emreulsy/nba-passing-analysis>

Acknowledgements

The author thanks dr. Arno Knobbe for the course and dataset, and Coach Bert Samson for the original research question.

References

1. Oliver, D.: Basketball on Paper: Rules and Tools for Performance Analysis. Potomac Books, Washington, D.C. (2004)
2. Atzmueller, M.: Subgroup discovery. *Wiley Interdisciplinary Reviews: Data Mining and Knowledge Discovery* **5**(1), 35–49 (2015)
3. Kubatko, J., Oliver, D., Pelton, K., Rosenbaum, D.T.: A starting point for analyzing basketball statistics. *Journal of Quantitative Analysis in Sports* **3**(3) (2007)
4. Sampaio, J., Lago, C., Drinkwater, E.J.: Explanations for the United States of America’s dominance in basketball at the Beijing Olympic Games. *Journal of Sports Sciences* **28**(2), 147–152 (2010)